# Metabolomics Data Matrix

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# Many variables in many samples

As in all *omic* experiments, a metabolomic assay measures a large number of **variables** (properties) in a (hopefully) reasonably large number of **samples**.

**Question**

... what are the variables ...

1. In a targeted metabolomics investigation?

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2. In an untargeted MS based metabolomics analysis?

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1. In a targeted metabolomics investigation?
2. In an untargeted MS based metabolomics analysis?
3. In an untargeted NMR based metabolomics assay?

# Number of variables | place your guess

- Targeted metabolomics investigation

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- Targeted metabolomics investigation
  - from 10 to 300

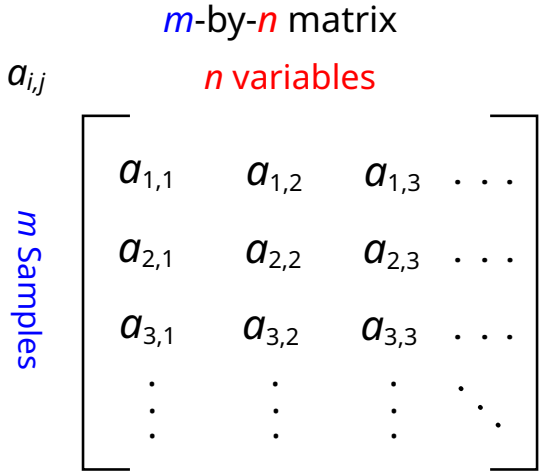
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- Targeted metabolomics investigation
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  - from 1000 to 30000

# Data Matrix


m-by-n matrix
$a_{i,j}$
n variables
m Samples
$a_{1,1}$ $a_{1,2}$ $a_{1,3}$ . . .
$a_{2,1}$ $a_{2,2}$ $a_{2,3}$ . . .
$a_{3,1}$ $a_{3,2}$ $a_{3,3}$ . . .

# The goal of data analysis

**Goal**

The objective of data analysis is to find **“organization”** inside the matrix

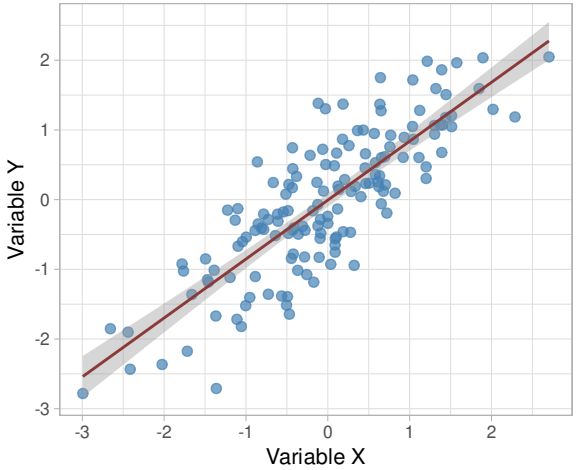
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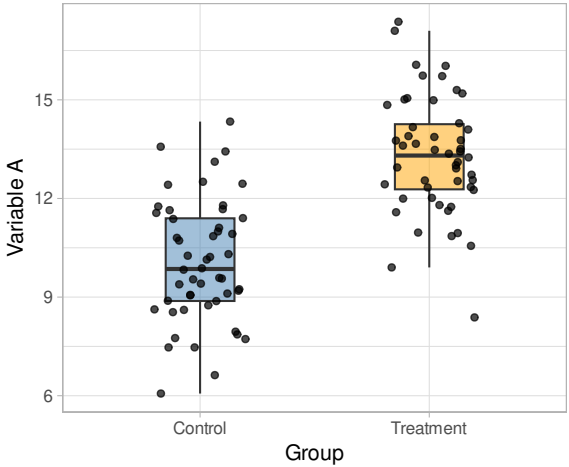
The objective of data analysis is to find **"organization"** inside the matrix

Our organization should also be statistically robust: if someone redid my experiment, the results should be the same

# Organization: Correlations



# Organization: Biomarkers

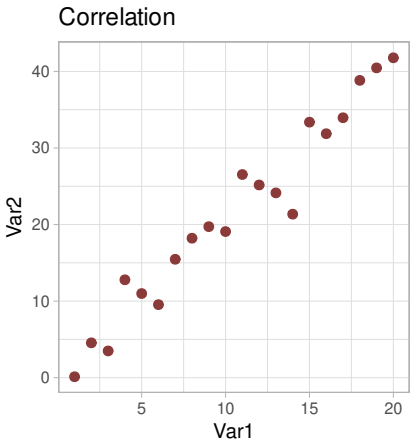


Section 1

# Data Matrices: facts

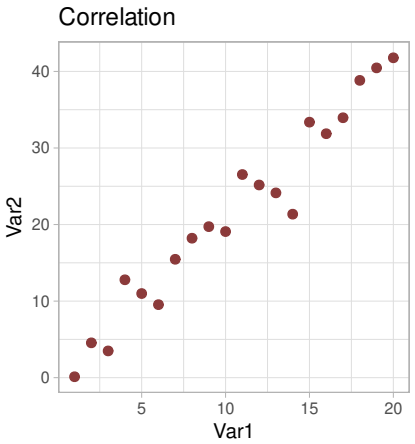
# Variables are not independent

- Biological relations



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- Biological relations
- Chemical/Analytical reasons



# Analytical vs biological correlation

**Important**

Analytical correlation is always stronger than the biological one!

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The correlation structure of your matrix is often not really informative about biology

# Rows (samples) are often dependent

The design of the study can result in a multilevel hierarchical structure of the samples which violates independence. In other words some samples are "by construction" associated and share something in common

**Examples**

1. Repeated measures of the same individual

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2. Subpopulations

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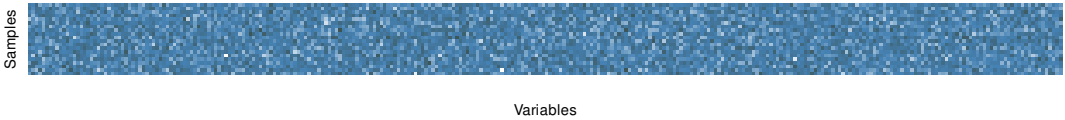
**Examples**

1. Repeated measures of the same individual
2. Subpopulations
3. Different sites, different days, ...

# Data Matrix size

In the typical "happy" statistical context, the number of variables we measure is smaller than the number of samples.

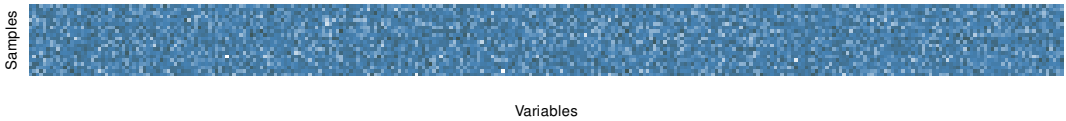
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FAT DATA MATRIX

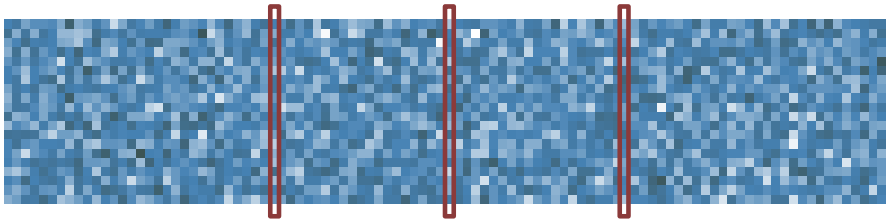
Section 2

# Analyzing data matrices

# Univariate approach

The **Univariate** approach considers each variable separately and applies "standard" statistical tools to spot the more **interesting** variables

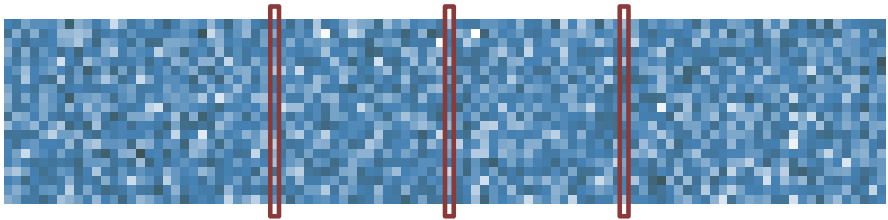
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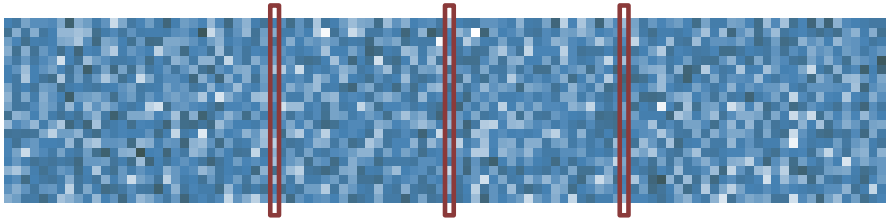
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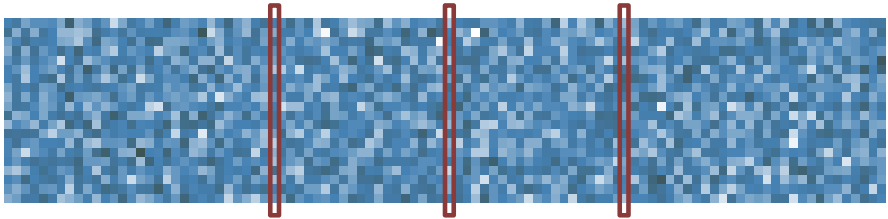
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3. ANOVAs



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1. Statistical testing
2. Linear modeling (`lm`, `glm`, ...)
3. ANOVAs
4. Hierarchical Modeling



# Univariate: pros and cons

**PRO**

1. Statistical modeling is there!
2. Interpretable by construction

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## PRO

1. Statistical modeling is there!
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## CONS

1. Multiple testing
2. The structure of the data creates redundancy
3. Assumptions for parametric (and non parametric) approaches are often not fulfilled

# Where is the problem: Multiple testing

- We measure many variables (features, metabolites) in the same set of samples

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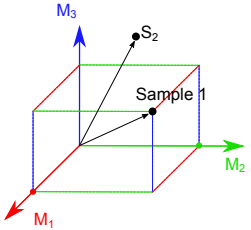
- We measure many variables (features, metabolites) in the same set of samples
- We run a battery of statistical tests looking for the significance of what we see in the individual variables
- **We ask ourselves if at least one test is significant in the overall set of tests**

We run individual tests, but we have a question about the full set of variables ...

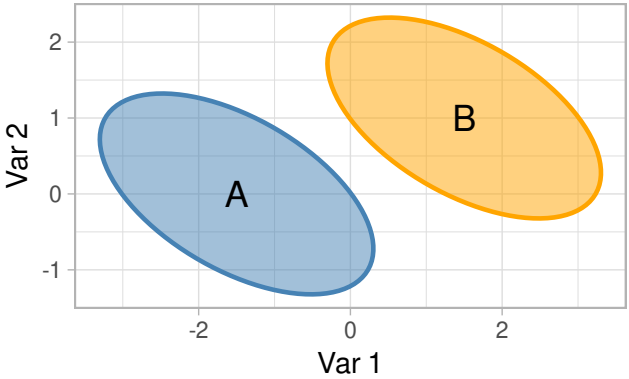
# Multivariate approach

Each sample is represented as a point in the multidimensional variable space. The dataset is a **cloud** of points in that space.

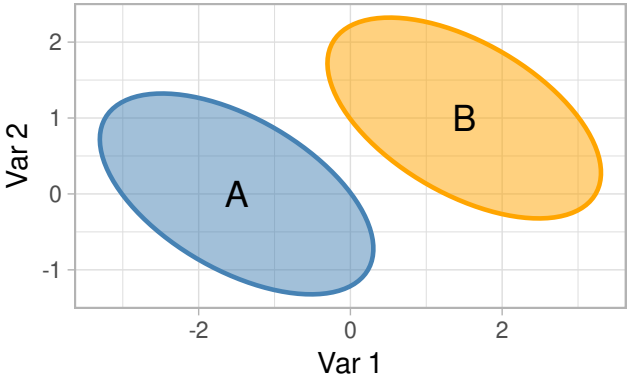
The size of the space equals the number of variables we are measuring. **Multivariate methods** (PCA, PLS, ASCA, ...) are able to exploit the correlation between the variables to highlight/extract the organization of the data



# Why multivariate ...



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Group separation is clearer in the 2D space ...

# Multivariate: pros and cons

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1. Potentially more powerful
2. Explicit use of variable correlation
3. No issues with multiple testing

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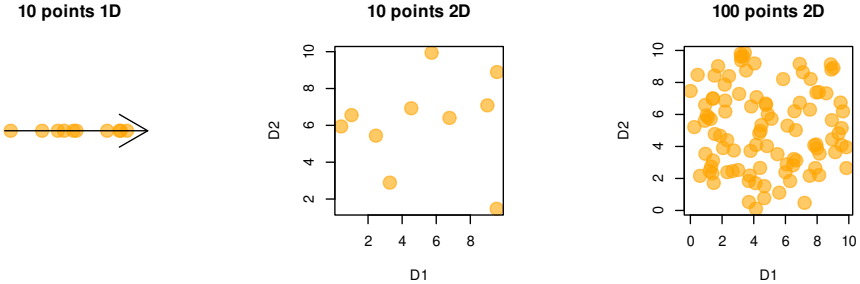
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## CONS

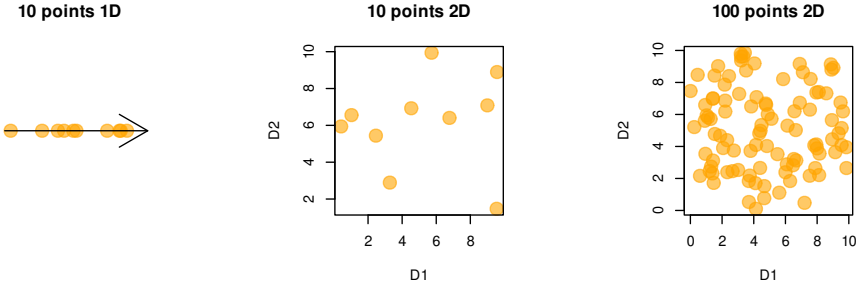
1. Chance correlations in fat matrices
2. Empty space
3. Difficult to embed hierarchical structure

# The problem: the curse of dimensionality



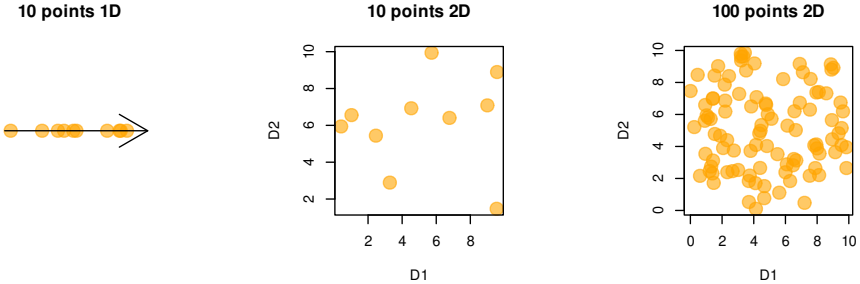

1. To fill the space, the number of points does not grow linearly with the number of dimensions

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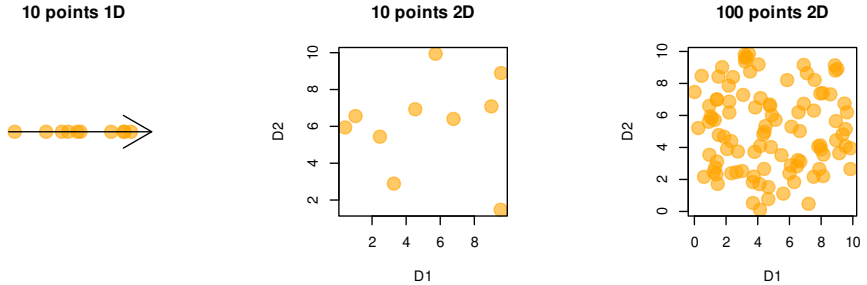

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Can you imagine 20 samples in 10000 dimensions?